

# An independent second proof of Said Duran's equal-height Collatz runs theorem

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## Abstract

Said Duran proved that arbitrarily long consecutive runs of positive integers have the same finite first hitting time of one under the Collatz map, with the same number of odd steps, above every prescribed lower bound. We give an independent second proof in the equivalent finite-pattern formulation. A direct induction synchronizes affine families on a common dyadic progression; a power-of-two endpoint then gives equal first hitting times. Lean 4 verifies the construction and its first-arrival conclusions. Priority for the existence theorem belongs to Said Duran. The proof selects translates of each finite pattern and does not establish convergence from an arbitrary prescribed starting integer.

## 1 Introduction and theorem

Garner's 1985 run-length conjecture [8], as recorded by Lagarias [12, entry 72, p. 29], asserts that arbitrarily long runs of consecutive integers with equal Collatz height exist. Said Duran [9, Theorem 1.1] proved this with equal odd-step counts and runs above every prescribed lower bound. His preprint is dated 27 September 2026; its Zenodo record was created on 28 September at 01:10:30 UTC, before this proof first appeared in its original combined manuscript. His Lean formalization is archived separately [10]. Priority for the existence theorem belongs to Said Duran.

This paper presents an independent second proof of the same result. Said Duran obtains synchronized runs from his density-one coalescence theorem [11]. Here a direct induction synchronizes affine families on a dyadic progression. Both arguments then choose a power-of-two common endpoint. We claim no stronger existence theorem. Throughout this paper,  $\mathbb{N}$  includes zero, and

$$T(x) = \begin{cases} x/2, & x \text{ even}, \\ (3x+1)/2, & x \text{ odd} \end{cases} \quad (x \in \mathbb{Z}_{>0}).$$

Write  $o_k(x)$  for the number of odd integers among  $x, T(x), \dots, T^{k-1}(x)$ , with  $o_0(x) = 0$ .

**Theorem 1** (Said Duran, equivalent finite-pattern form). *For every finite set  $F \subset \mathbb{N}$  and every  $L \in \mathbb{N}$ , there are integers  $x > \max\{1, L\}$  and  $H, R \in \mathbb{N}$  such that, for every  $n \in F$ ,*

$$T^H(x+n) = 1, \quad o_H(x+n) = R, \quad T^j(x+n) > 1 \quad (0 \leq j < H). \quad (1)$$

*In particular, every prescribed finite pattern has arbitrarily large translates whose members have the same finite first hitting time of one and the same number of odd steps before that time.*

Taking  $F = \{0, 1, \dots, \ell - 1\}$  gives arbitrarily long intervals with these properties. Conversely, a nonempty finite  $F$  is contained in  $\{0, \dots, \max F\}$ , so a run of length  $1 + \max F$  gives its finite-pattern instance; the empty case is vacuous. For example, the starts 28, 29, 30 all reach 20 after seven steps and first reach one after thirteen steps, each having used five odd steps. Earlier studies of equal-height pairs and consecutive runs include Garner [8] and Gao [14]; LaDue [16] studies clusters with equal total stopping times. Garner's separate pair-characterization conjecture, refuted by Elia and Tucker [13], is distinct from the run-length conjecture considered here. The final power-of-two step has a classical predecessor in the repetition lemma: a known equal-height run

can be shifted to produce infinitely many runs of the same length. Penning's 1989 observation, recorded in the comments to Problem 1370 in *Crux Mathematicorum* [5], uses an affine common value and chooses the shift to make it a power of two. Gao [14, Lemma 1], crediting Penning, and Wu [6] proved repetition results. Our proof first constructs a synchronized finite pattern and then applies the same power-of-two principle. The classical fact that powers of two exhaust the units modulo powers of three also underlies the back-tracing argument in Monks [15, Lemma 3.4].

A closely related earlier computational construction appears in Sawicki's 2024 Math StackExchange answer [7]. He considers starts  $n2^a + b$ , reduces their trajectories to finitely many affine values  $n3^c + d$ , and searches for a parameter giving equal remaining heights. His procedure first computes the small starting orbits and explicitly leaves success of the final parameter search unproved. The affine-family synchronization below supplies an unconditional existence argument for every finite pattern.

## 2 Parity classes and prescribed gaps

The following is the classical parity-class formula of Terras [3] and Everett [4]. We include its elementary proof to fix the notation.

**Lemma 2.** For  $c \in \mathbb{Z}_{>0}$ ,  $k, t \in \mathbb{N}$ , and  $0 \leq j \leq k$ ,

$$T^j(c + 2^k t) = T^j(c) + 2^{k-j} 3^{o_j(c)} t, \quad (2)$$

$$o_j(c + 2^k t) = o_j(c). \quad (3)$$

*Proof.* The assertions hold at  $j = 0$ . If  $j < k$ , the difference in (2) is even, so the two current integers have the same parity. An even step divides the difference by two; an odd step multiplies it by  $3/2$  and adds one to both odd counts. This proves the induction.  $\square$

In particular, the first  $k$  parities are constant on each positive residue class modulo  $2^k$ . If a current value is  $A + 3^r t$ , either parity can be prescribed by restricting  $t$  to  $\varepsilon + 2u$ , where  $\varepsilon \in \{0, 1\}$ . After the prescribed actual step, the coefficient of  $u$  is  $3^r$  or  $3^{r+1}$  according as that step is even or odd. Thus this prescription can be repeated, and each additional step doubles the parameter modulus.

**Lemma 3.** For every  $n \in \mathbb{N}$ , there are  $k \in \mathbb{N}$  and  $a \in \mathbb{Z}_{>0}$  such that, for every  $t \in \mathbb{N}$ ,

$$T^k(a + 2^k t) = T^k(a + 2^k t + n),$$

$$o_k(a + 2^k t) = o_k(a + 2^k t + n).$$

*Proof.* We use strong induction on  $n$ . For  $n = 0$ , take  $k = 0$  and  $a = 1$ . For  $n = 1$ , the two trajectories

$$4 + 8t \mapsto 2 + 4t \mapsto 1 + 2t \mapsto 2 + 3t,$$

$$5 + 8t \mapsto 8 + 12t \mapsto 4 + 6t \mapsto 2 + 3t$$

each have one odd step. This gives  $k = 3$  and  $a = 4$ .

If  $n = 2m > 0$ , use the progression  $a + 2^k t$  for  $m$  supplied by induction. The starts  $2a + 2^{k+1}t$  and  $2a + 2^{k+1}t + 2m$  take one even step to that pair. They therefore coalesce after  $k + 1$  steps, with equal odd counts.

It remains to treat odd  $n > 1$ . Choose  $m, c, d$  by

| $n \bmod 4$ | $m$          | $c$ | $d$ |
|-------------|--------------|-----|-----|
| 1           | $(3n + 1)/4$ | 1   | 2   |
| 3           | $(3n - 1)/4$ | 2   | 1   |

so that  $0 < m < n$  and  $3n + d = 4m + c$ . Let  $a + 2^k t$  be the progression for  $m$ . Because  $2^k$  is invertible modulo three, some  $q \in \{0, 1, 2\}$  makes  $4(a + 2^k q) - c$  divisible by three. The integer

$$a' = \frac{4(a + 2^k q) - c}{3}$$

is positive. For  $X = a' + 2^{k+2}t$  and  $Y = a + 2^k(q + 3t)$ , we have

$$3X + c = 4Y, \quad 3(X + n) + d = 4(Y + m).$$

If  $3z + 1 = 4w$ , then  $z \equiv 1 \pmod{4}$  and its next two steps are odd, even, with endpoint  $w$ . If  $3z + 2 = 4w$ , then  $z \equiv 2 \pmod{4}$  and its next two steps are even, odd, again with endpoint  $w$ . Hence  $X$  and  $X + n$  reach  $Y$  and  $Y + m$  after two steps, each having used one odd step. The induction hypothesis at parameter  $q + 3t$  finishes the construction.  $\square$

### 3 Synchronizing affine families

To add an offset to a collection that already coalesces, we must synchronize its current trajectory with the common trajectory of the collection. Their coefficients in the remaining parameter can be different powers of three.

**Lemma 4.** *Let  $A, B \in \mathbb{Z}_{>0}$  and  $r, s \in \mathbb{N}$ . There exist  $\ell, t_0, Q \in \mathbb{N}$  and  $C \in \mathbb{Z}_{>0}$  such that, for every  $v \in \mathbb{N}$ ,*

$$T^\ell(A + 3^r(t_0 + 2^\ell v)) = T^\ell(B + 3^s(t_0 + 2^\ell v)) = C + 3^Q v. \quad (4)$$

*The two length- $\ell$  trajectories have fixed odd counts  $u, w$  satisfying  $r + u = s + w = Q$ .*

*Proof.* Exchange the families if necessary so that  $r \leq s$ . Initially their current values satisfy

$$Y = 3^d X + D, \quad d = s - r, \quad D = B - 3^d A \in \mathbb{Z}.$$

At every subsequent stage the coefficient of the remaining parameter in  $X$  is a power of three and is therefore odd. We may prescribe its next parity, as above. The displayed relation determines the parity of  $Y$ . When  $d > 0$ , the following steps reduce the relation to one with a smaller exponent.

If  $D \neq 0$  is even, prescribe  $X$  even. Both values take an even step, and

$$Y' = 3^d X' + D/2.$$

Repeating this finitely many times makes  $D$  odd. This also applies to negative  $D$ , by dividing out the power of two in  $|D|$ . If  $D = 0$ , prescribe  $X$  odd. Both values take an odd step, and

$$Y' = 3^d X' + \frac{1 - 3^d}{2}.$$

The new constant is nonzero because  $d > 0$ , so the preceding halving procedure makes it odd. Once  $D$  is odd, prescribe  $X$  odd. Then  $Y$  is even, and

$$X' = \frac{3X + 1}{2}, \quad Y' = \frac{Y}{2}, \quad Y' = 3^{d-1} X' + \frac{D - 3^{d-1}}{2}. \quad (5)$$

The last constant is an integer and the exponent has decreased. There are finitely many steps at each fixed positive  $d$  and finitely many decreases of  $d$ , so eventually  $d = 0$ .

The current values now have the same coefficient in the remaining parameter and differ by a constant integer  $D$ . If  $D = 0$ , they are equal already. Otherwise, apply Lemma 3 with gap  $|D|$

to the smaller current value. Suppose its required progression is  $a + 2^k z$ . The smaller value has the form  $E + 3^q u$ , so the congruence

$$E + 3^q u_0 \equiv a \pmod{2^k}$$

has a solution. Choose a nonnegative representative  $u_0$  large enough that  $E + 3^q u_0 \geq a$ . On the parameter class  $u = u_0 + 2^k v$ , the smaller value belongs to the positive tail of the required progression for every  $v \geq 0$ . The next  $k$  steps therefore make the two values equal.

Each parameter restriction has modulus two for each added actual step. Their composition has the form  $t = t_0 + 2^\ell v$ , where  $\ell$  is the common total number of steps. All bases and current values remain positive. By Lemma 2, the resulting odd counts  $u, w$  are independent of  $v$ , and the two final affine coefficients are  $3^{r+u}$  and  $3^{s+w}$ . Equality at  $v = 0$  and  $v = 1$  forces these coefficients to agree. This proves (4) and  $r + u = s + w = Q$ .  $\square$

**Proposition 5.** *For every finite  $F \subset \mathbb{N}$ , there are  $K, R \in \mathbb{N}$  and  $a, b \in \mathbb{Z}_{>0}$  such that, for every  $n \in F$  and  $t \in \mathbb{N}$ ,*

$$T^K(a + 2^K t + n) = b + 3^R t, \quad o_K(a + 2^K t + n) = R. \quad (6)$$

*Proof.* Enlarge  $F$  to include zero, and enumerate its offsets starting with zero. For the initial singleton take  $K = R = 0$  and  $a = b = 1$ . Suppose the assertion holds for the offsets already included. Their common current value on  $a + 2^K t$  is  $U(t) = b + 3^R t$ . For the next offset  $n$ , Lemma 2 gives

$$T^K(a + 2^K t + n) = d + 3^S t, \quad d = T^K(a + n) > 0, \quad S = o_K(a + n).$$

Apply Lemma 4 to these two affine families. On a class  $t = t_0 + 2^\ell v$ , they have the same image after  $\ell$  additional steps. Every previously included trajectory starts those steps at exactly  $U(t)$ , so all previous equalities are preserved.

The new starting progression is

$$(a + 2^K t_0) + 2^{K+\ell} v.$$

Its modulus agrees with the new common time  $K + \ell$ . If the two added trajectories use  $u, w$  odd steps, the last assertion of Lemma 4 gives  $R + u = S + w$ . Thus all total odd counts agree as well. The new endpoint has the required form  $b' + 3^{R+u} v$ . Finite induction proves the proposition.  $\square$

## 4 Specializing the endpoint to a power of two

**Lemma 6.** *Let  $b \in \mathbb{Z}_{>0}$  with  $3 \nmid b$ , and let  $R \in \mathbb{N}$ . There are arbitrarily large  $t \in \mathbb{N}$  for which  $b + 3^R t$  is a power of two.*

*Proof.* We give the residue argument explicitly. For  $m \geq 1$ ,

$$4^{3^{m-1}} \equiv 1 + 3^m \pmod{3^{m+1}}. \quad (7)$$

The case  $m = 1$  is immediate. Cubing a number of the form  $1 + 3^m(1 + 3c)$  proves the next case, since the terms beyond the linear term are divisible by  $3^{m+2}$ .

Powers of two represent both units modulo three. Suppose  $2^s \equiv b \pmod{3^m}$ . Multiplying  $2^s$  by  $(4^{3^{m-1}})^j$ , for  $j = 0, 1, 2$ , changes its residue modulo  $3^{m+1}$  by  $j2^s 3^m$ , by (7). Since  $3 \nmid 2^s$ , these are the three possible lifts of its residue modulo  $3^m$ . One therefore represents  $b$  modulo  $3^{m+1}$ . Induction gives  $2^s \equiv b \pmod{3^R}$  for every  $R \geq 1$ ; the case  $R = 0$  is immediate.

The same identity gives  $2^{2 \cdot 3^R} \equiv 1 \pmod{3^R}$ . We may consequently increase  $s$  by arbitrary multiples of  $2 \cdot 3^R$  while preserving the residue. For all sufficiently large such exponents,

$$t = \frac{2^s - b}{3^R}$$

is a nonnegative integer, and these values of  $t$  are unbounded.  $\square$

*Proof of Theorem 1.* Apply Proposition 5 to  $F \cup \{0, 1\}$ , obtaining  $K, R, a, b$ . We first show  $3 \nmid b$ . An odd step has output congruent to two modulo three. Thus a current value divisible by three has only an even predecessor, which is also divisible by three. If  $3 \mid b$ , every step in any length- $K$  trajectory ending at  $b$  would be even, and its initial value would equal  $2^K b$ . The distinct starts  $a$  and  $a + 1$  both end at  $b$ , a contradiction.

By Lemma 6, choose  $t \geq 1$  as large as necessary, with  $b + 3^R t = 2^s$ , and put  $x = a + 2^K t$ . The common endpoint after  $K$  steps is  $2^s$ . After  $s$  further even steps, all trajectories reach one, with total odd count  $R$ . Taking  $t$  large ensures  $x > \max\{1, L\}$ .

It remains to check that this is the first visit to one. For  $j < K$ , Lemma 2 gives

$$T^j(x + n) = T^j(a + n) + 2^{K-j} 3^{o_j(a+n)} t > 1.$$

For  $j = K + q < K + s$ , the value is  $2^{s-q} > 1$ . Hence  $H = K + s$  satisfies all of (1).  $\square$

For the unshortened map that sends odd  $x$  to  $3x + 1$ , each of the  $R$  odd steps above expands to an odd step followed by an even step. The corresponding first hitting time is therefore  $H + R$ , also common to the whole pattern. The inserted intermediate values are greater than one, so first arrival is preserved.

The translated patterns in Theorem 1 are selected by the construction. The theorem makes no assertion that an arbitrary prescribed starting integer reaches one or enters one of these patterns. In particular, it does not prove the Collatz conjecture.

## 5 Formal verification and scope

The formalization uses Lean 4 and mathlib [1, 2]. The declarations use namespace

`CollatzPositiveProgression.`

The map `step` is the shortcut Collatz map on natural numbers; `iterate` and `oddCount` are defined recursively. The entry point

`formal/FinitePatternConvergence.lean`

imports the finite-pattern coalescence theorem and the power-of-two specialization. Its declarations

`every_finite_pattern_has_equal_first_hitting_times`  
`arbitrarily_long_consecutive_equal_first_hitting_times`

prove Theorem 1 and its consecutive-run corollary. They quantify over actual iterates and explicitly require every earlier iterate to exceed one. Appendix A reproduces the definitions and exact principal theorem types.

The checked declarations report only `propext`, `Classical.choice`, and `Quot.sound`; no additional axiom or proof placeholder is used. They are included in the retained combined fresh build of 137 local modules, whose audit checks 633 public theorems and lemmas and 93 supporting definitions. That combined artifact also contains separate matrix-interpretation results not used here. The checked environment is Lean 4.27.0 with mathlib revision

a3a10db0e9d66acbebf76c5e6a135066525ac900.

The source `formal/FinitePatternConvergence.lean` has SHA-256

74ccd96f004198fbd04c6a5283022cb78414fd0351f49bc4c0ad61fc1c30ffb5.

The source, verification script, retained compiler output, and claim-to-declaration map are available at

<https://github.com/x1xhlol/collatz-two-coordinate-obstruction>.

From the artifact root, reproduction uses

```
python3 verify.py \
  --mathlib-root /path/to/pinned/mathlib \
  --build-root /path/to/temporary/builds \
  --report /path/to/fresh-check.json
```

with a new output path, preserving the retained report `verification/rebuild.json`. The claim map is `paper/lean-claims.md`. Lean’s kernel checks the proof terms against the displayed definitions and the stated foundational axioms. These theorems construct suitable translates of finite patterns; they do not assert convergence for an arbitrary fixed starting value.

## Acknowledgments

I thank O. J. Said Duran for making his earlier theorem and formalization available. The Lean development uses `mathlib`, and I thank the developers and contributors of both projects.

## Use of AI

OpenAI Codex was used extensively for mathematical exploration, development of the Lean proofs and verification scripts, and preparation and revision of this manuscript. The proof has the accompanying Lean artifact. Automated review does not constitute external peer review. This disclosure does not imply review or endorsement by Said Duran.

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## A Exact Lean declarations

The blocks below reproduce the checked source verbatim, with the original line numbers. The theorem blocks end at their types; their proof terms are omitted. Source paths are relative to the repository root.

The following declarations use namespace `CollatzPositiveProgression`. Natural numbers include zero. The constructed starts are positive. Here `oddCount k x` counts odd inputs in the first  $k$  steps. The first-hitting-time theorem requires every earlier value to exceed one.

### The map, its iterates, and its odd-step count.

`formal/PositiveProgressionCoalescence.lean`, lines 9–18.

```

9 def step (x : ℕ) : ℕ :=
10   if x % 2 = 0 then x / 2 else (3 * x + 1) / 2
11
12 def iterate : ℕ → ℕ → ℕ
13   | 0, x => x
14   | k + 1, x => iterate k (step x)
15
16 def oddCount : ℕ → ℕ → ℕ
17   | 0, _ => 0
18   | k + 1, x => x % 2 + oddCount k (step x)

```

### Coalescence on a common dyadic progression.

`formal/FinitePatternCoalescence.lean`, lines 50–53.

```

50 theorem every_finite_pattern_coalesces (F : Finset ℕ) :
51   ∃ K R a b : ℕ, 0 < a ∧ 0 < b ∧ ∀ n ∈ F, ∀ t : ℕ,
52     iterate K (a + 2 ^ K * t + n) = b + 3 ^ R * t ∧
53     oddCount K (a + 2 ^ K * t + n) = R

```

### Equal finite first hitting times for any finite pattern.

`formal/FinitePatternConvergence.lean`, lines 8–12.

```

8 theorem every_finite_pattern_has_equal_first_hitting_times
9   (F : Finset ℕ) (lower : ℕ) :
10   ∃ x H R : ℕ, lower < x ∧ 1 < x ∧ ∀ n ∈ F,
11     iterate H (x + n) = 1 ∧ oddCount H (x + n) = R ∧
12     ∀ j < H, 1 < iterate j (x + n)

```

### Consecutive runs of every prescribed length.

`formal/FinitePatternConvergence.lean`, lines 32–36.

```

32 theorem arbitrarily_long_consecutive_equal_first_hitting_times
33   (length lower : ℕ) :
34   ∃ x H R : ℕ, lower < x ∧ 1 < x ∧ ∀ n < length,
35     iterate H (x + n) = 1 ∧ oddCount H (x + n) = R ∧
36     ∀ j < H, 1 < iterate j (x + n)

```